

Title: Spectral Analysis

Category: Analysis

1. Objective: To decode the "State of Mind" of the user. Raw EEG signals are a complex mix of many different waves. This method breaks the signal down into its component frequencies to determine which brain waves are dominant (e.g., Alpha for relaxation, Beta for focus).

2. How it Works (Mathematical Concept) (Using SciPy Library)

The algorithm uses Welch's Method to estimate the Power Spectral Density (PSD).

- **Step 1: Windowing**

Welch's method splits the signal into overlapping segments (windows). This reduces noise and provides a smoother, more accurate spectrum.

- **Step 2: Band Integration (Area Under the Curve)**

We calculate the "Relative Power" of each specific band by integrating the area under the PSD curve for that frequency range.

- **Formula:**

$$P_{band} = \int_{f_{low}}^{f_{high}} PSD(f) df$$

- **Relative Power:**

$$R_{band} = \frac{P_{band}}{P_{total}}$$

3. The Frequency Bands (Biological Meaning)

Instead of raw numbers, we analyze the biological significance of the 5 bands extracted:

- **Delta (0.5 - 4 Hz):** Associated with deep sleep or unconsciousness. High Delta in awake subjects often indicates artifacts (eye blinks).
- **Theta (4 - 8 Hz):** Associated with drowsiness, meditation, or "zoning out."
- **Alpha (8 - 13 Hz):** The "Idle State" of the brain. Dominant during relaxation with eyes closed.
- **Beta (13 - 30 Hz):** The "Active State." Dominant during problem-solving, focus, or anxiety.
- **Gamma (30 - 45 Hz):** Associated with high-level cognitive processing and binding information.

(Frequency band definitions adopted from the International Federation of Clinical Neurophysiology (IFCN) standards and Niedermeyer's Electroencephalography (7th Ed).)

4. Visual Result

(Insert Figure 1: Power Distribution Chart)

Description: The visualization displays the "Relative Power" of each band.

